



TRANSDUCER THEORY

FINAL PROJECT



By
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Professor:
Dr. Christopher Bennett

INITIAL CONCEPT

I found an old Acoustic MK260 Bass Amp Head

- **The Amplifier** works like new, aside from a few scratches on the exterior.
- **The Driver** stopped working in 2017.
- **The Cabinet** was unusual...
 - Top heavy
 - The bottom half of the cabinet was empty
 - Used as furniture and then tossed in 2022

The question is, can I create a new passive cabinet for this amplifier that works with a 5 string bass?



CONSTRAINTS

Amplifier

- 100W RMS
- ~200W Peak
- 4Ω
- Enclosure Length $\geq 13''$
- Enclosure Depth $\geq 8.5''$

5 String Bass

- E1: 41.2 Hz (Standard tuning 4 string)
- B0: 31.9 Hz (Standard tuning 5 string)
- A0: 27.5 Hz (Drop A tuning)

Driver

- Enclosure Length > Driver Diameter
- Enclosure Width > Driver Diameter

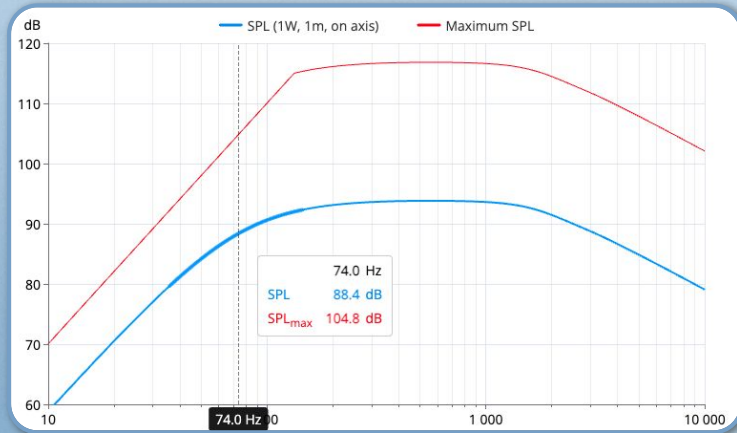


DRIVER SELECTION



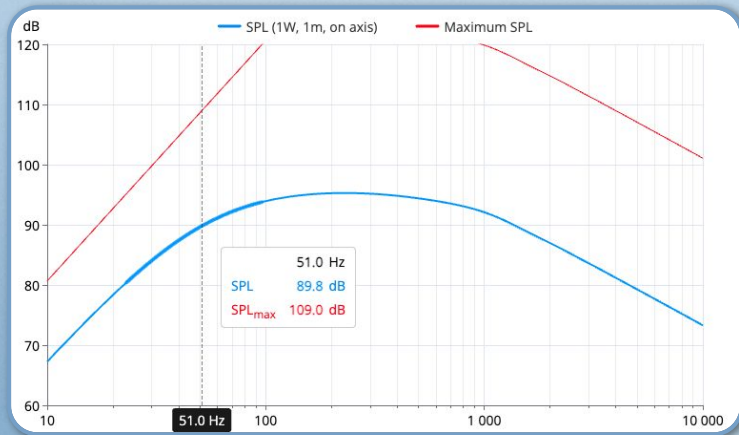
DRIVER SELECTION

- ✗ Celestion 10" BL10 100W 8 Ω
 - Very high f_s
 - No headroom



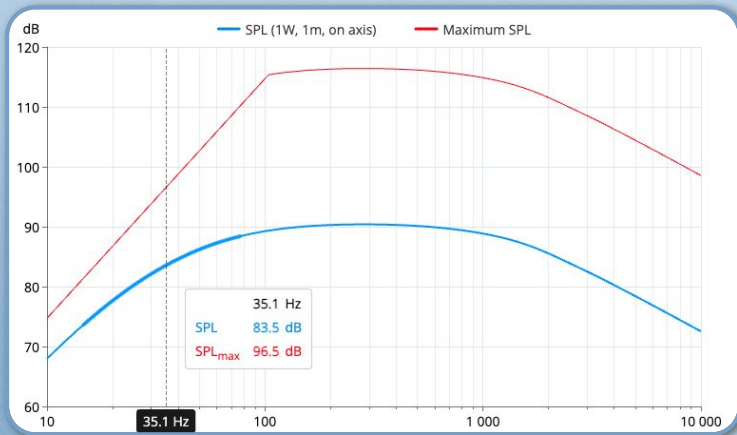
DRIVER SELECTION

- ✗ Eminence CA154 15" 300W 4Ω
 - Loud 🙄
 - Average f_s
 - Larger/heavier than original driver



DRIVER SELECTION

- ✓ Eminence 10" BP102 400W 8Ω
 - Higher wattage
 - Ideal f_s
 - Ideal size



DRIVER SELECTION

✓ Eminence 10" BP102 400W 8Ω

Key T/S Parameters:

- Resonant Frequency: 35 Hz
- DC Resistance: 2.7 Ω
- Electromagnetic Q: 0.4
- Mechanical Q: 10.61
- Total Q: 0.38
- Compliance Equivalent Volume: 103.8 liters
- Maximum Linear Excursion: 6.2 mm
- Surface Area of Cone: 344.9 cm²
- Voice Coil Inductance: 0.68 mH

Recommended Enclosure Specifications:

- Sealed Volume: 14–31 liters
- Vented Volume: 24–76 liters
- Driver Displacement: 1.17 liters

Physical Properties:

- Nominal Diameter: 10 inch (254mm)
- Magnet Weight: 38 oz.
- Overall Diameter: 10.08 inch (25mm)
- Baffle Cutout Diameter: 9.18 inch (233.2 mm)
- Weight: 8.9 lbs (4.04 kg)



Metric used for design and SI used for build.

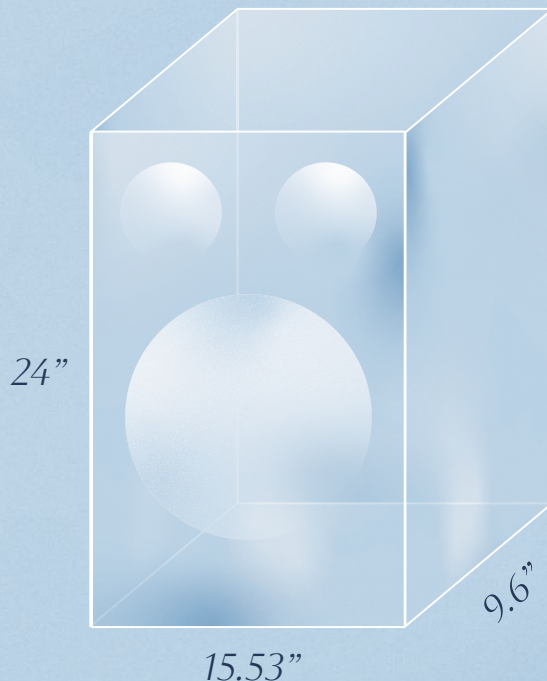
ENCLOSURE DESIGN

- A ported enclosure was selected.
- The enclosure was designed with an internal volume of 62 liters, to compromise between flat frequency response and realistic size
 - Port Diameter: 65 mm
 - Cross sectional port area: 6608 mm²
 - Port Length: ~160 mm
 - With flush end correction: ~130 mm

CAVEATS

- Volume was slightly reduced for portable enclosure
- Golden ratio dimensions calculated but slightly adjusted so the height is 24", not 24.7"
- Two ports chosen for ideal port lengths

INTERNAL DIMENSIONS



PREDICTED PERFORMANCE

Res. Freq: 39.1 Hz

R_e : 3.2 Ω

SPL: 118.5 dB

Sensitivity: 92.3 dB (1-W/1-m)

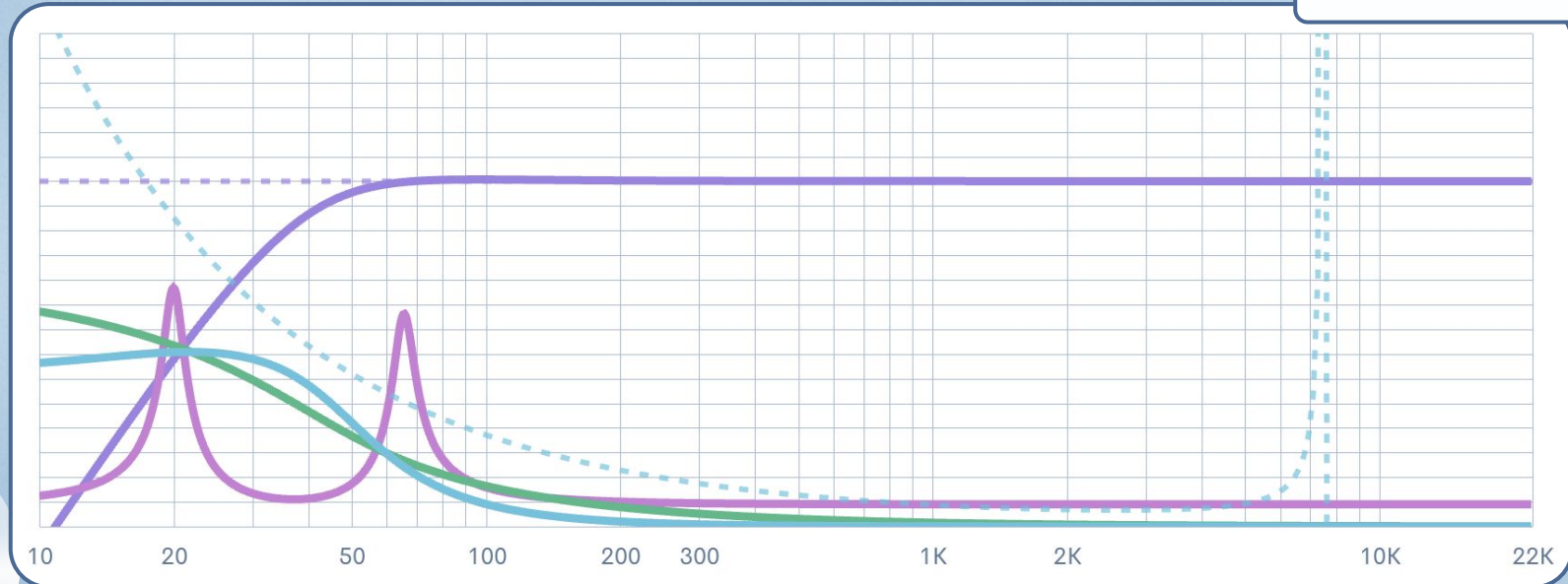
Res. Gain: <1 dB

Z_{MAX} : 29 Ω

F3: 42.1 dB

Efficiency (η): 1.08%

All Graphs



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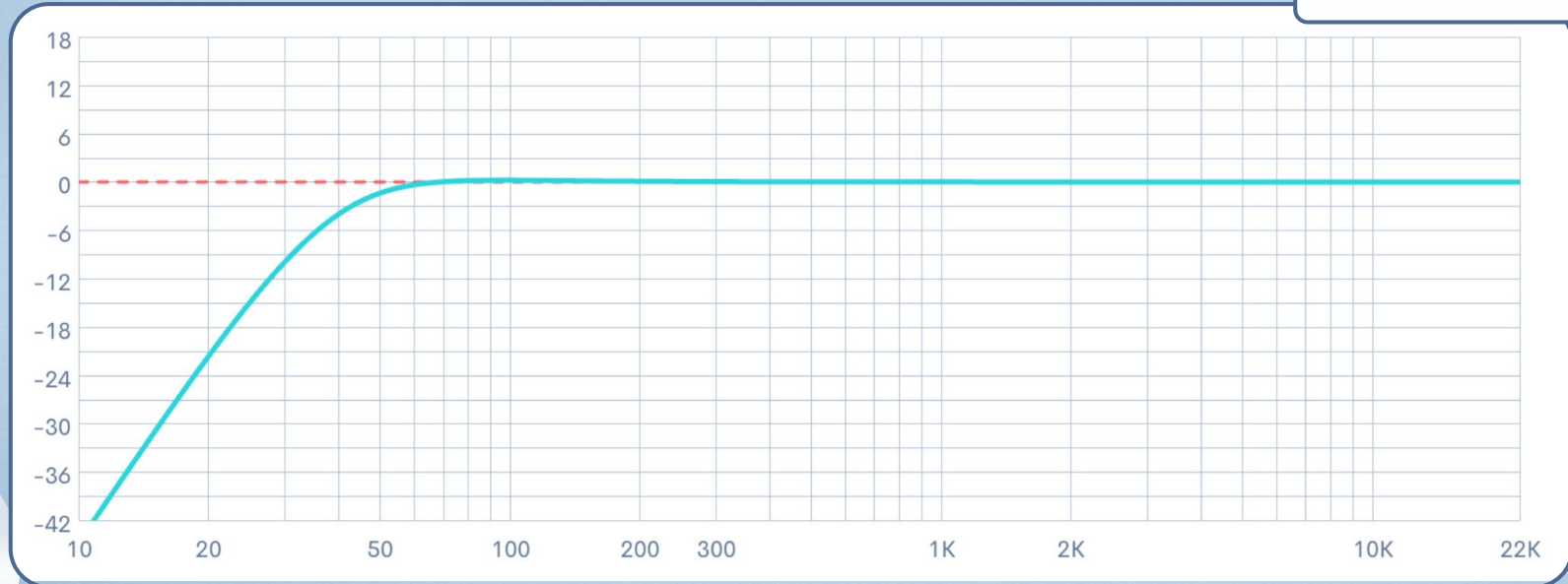
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Transfer Function (dB)



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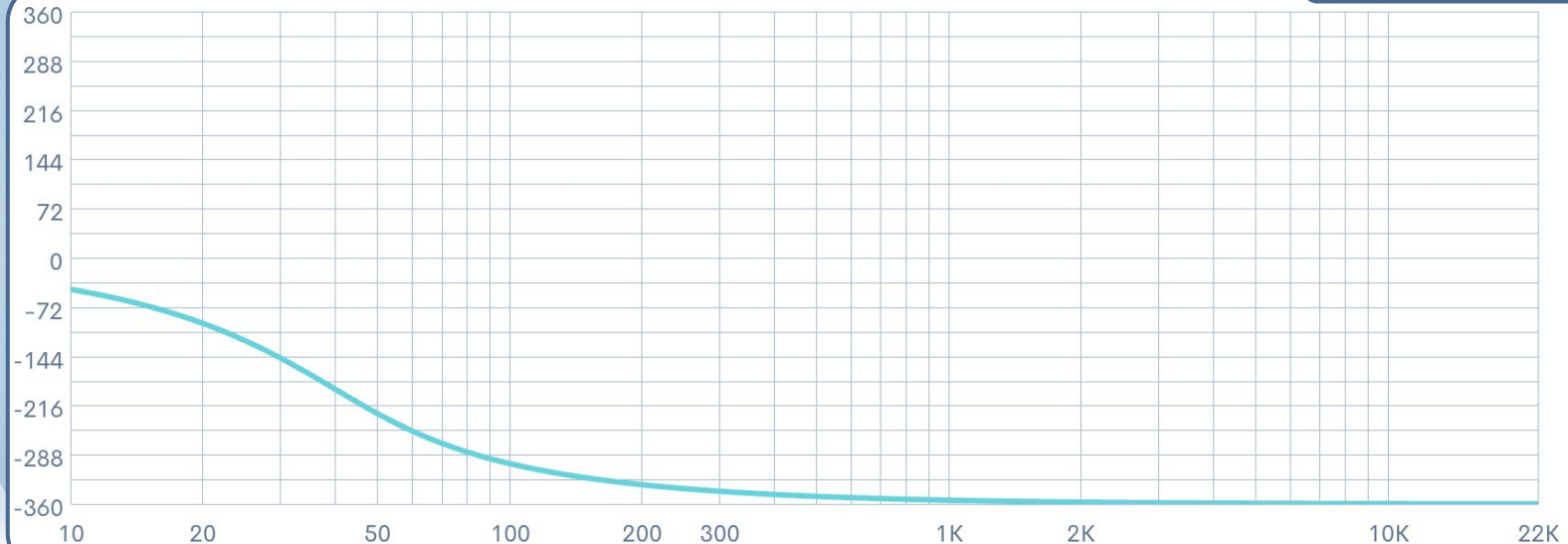
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Phase (°)



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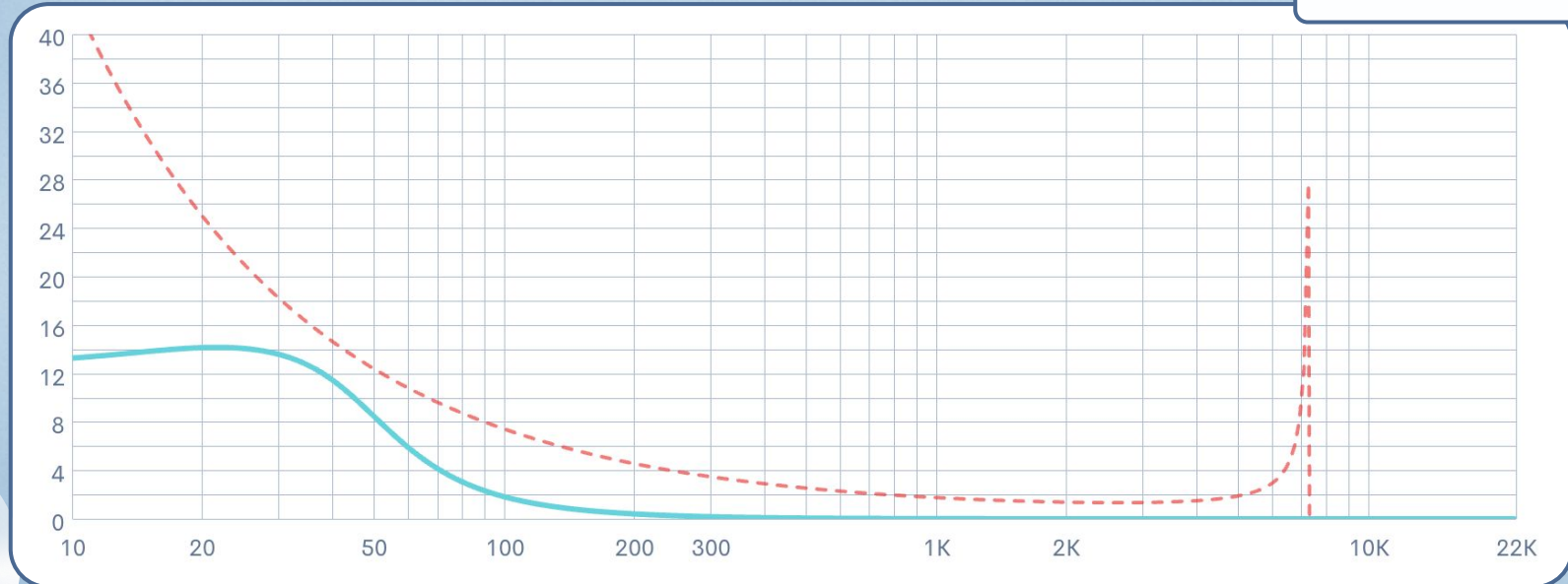
Res. Gain: <1 dB

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Group Delay (ms)



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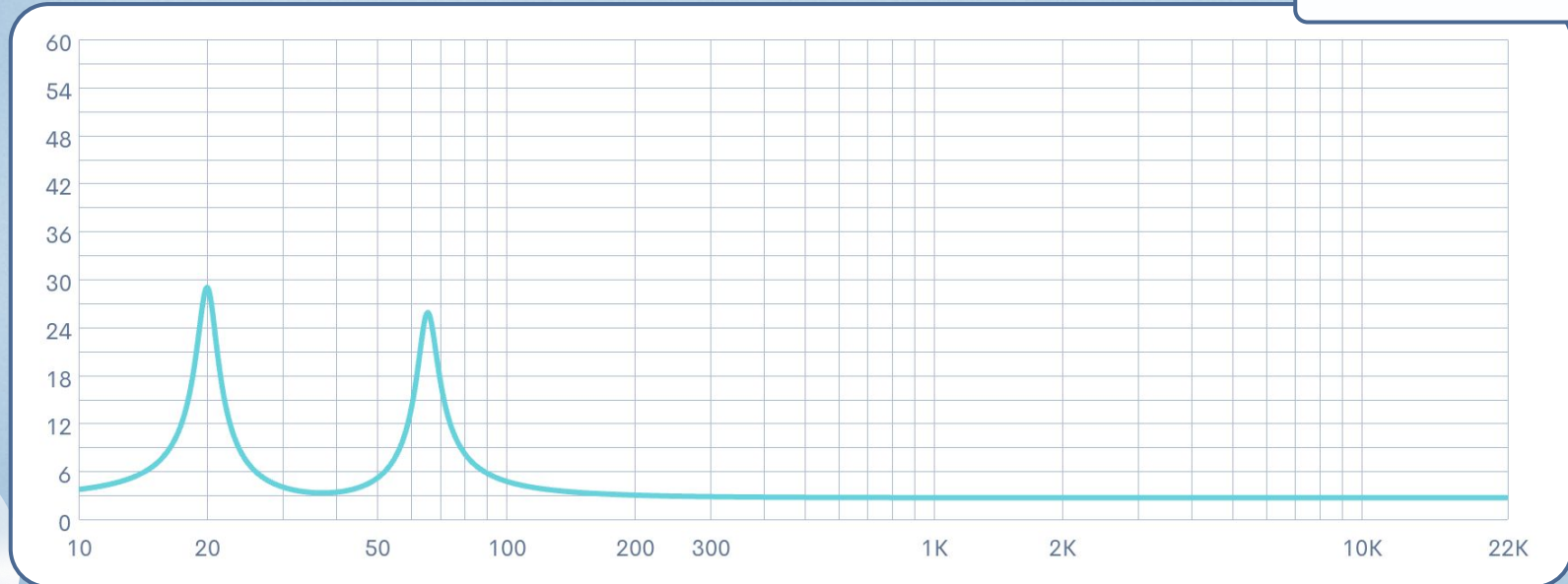
Res. Gain: <1 dB

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Impedance (Ω)



ENCLOSURE BUILD

COMPONENTS

- Birch Plywood
- Plastic Ports
- Driver
- Cabinet Terminal

ADDITIONS

- Rubber Feet
- Handle



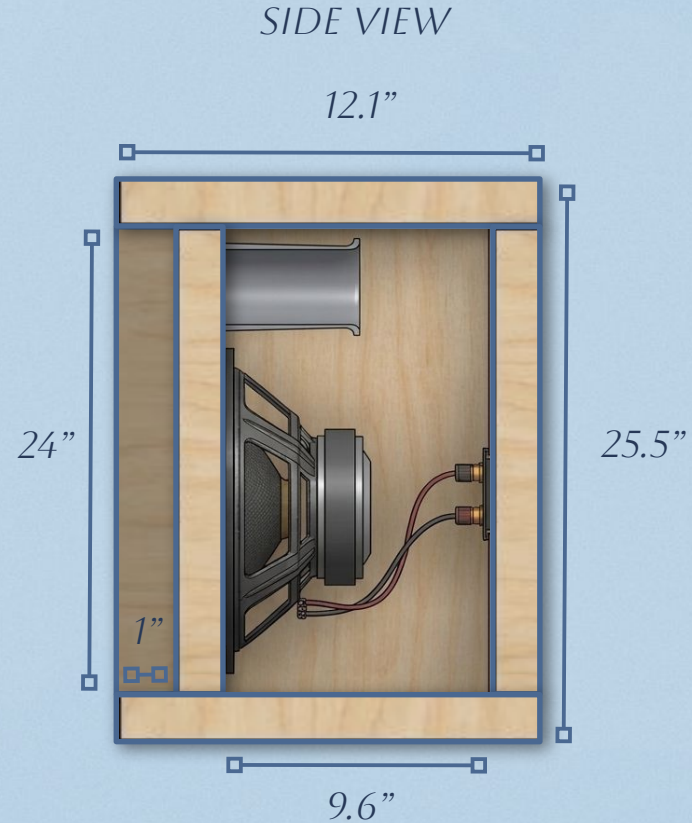
ENCLOSURE BUILD

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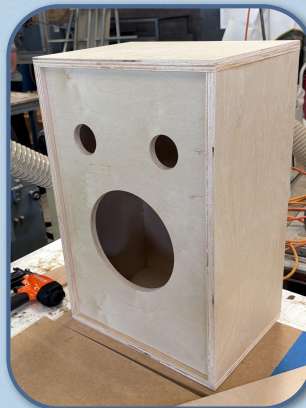
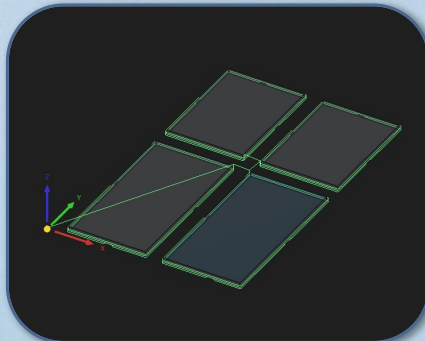
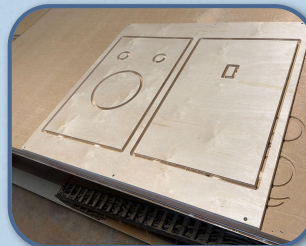
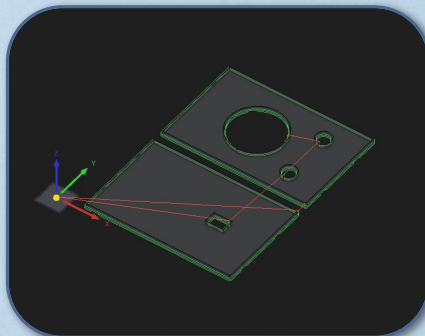
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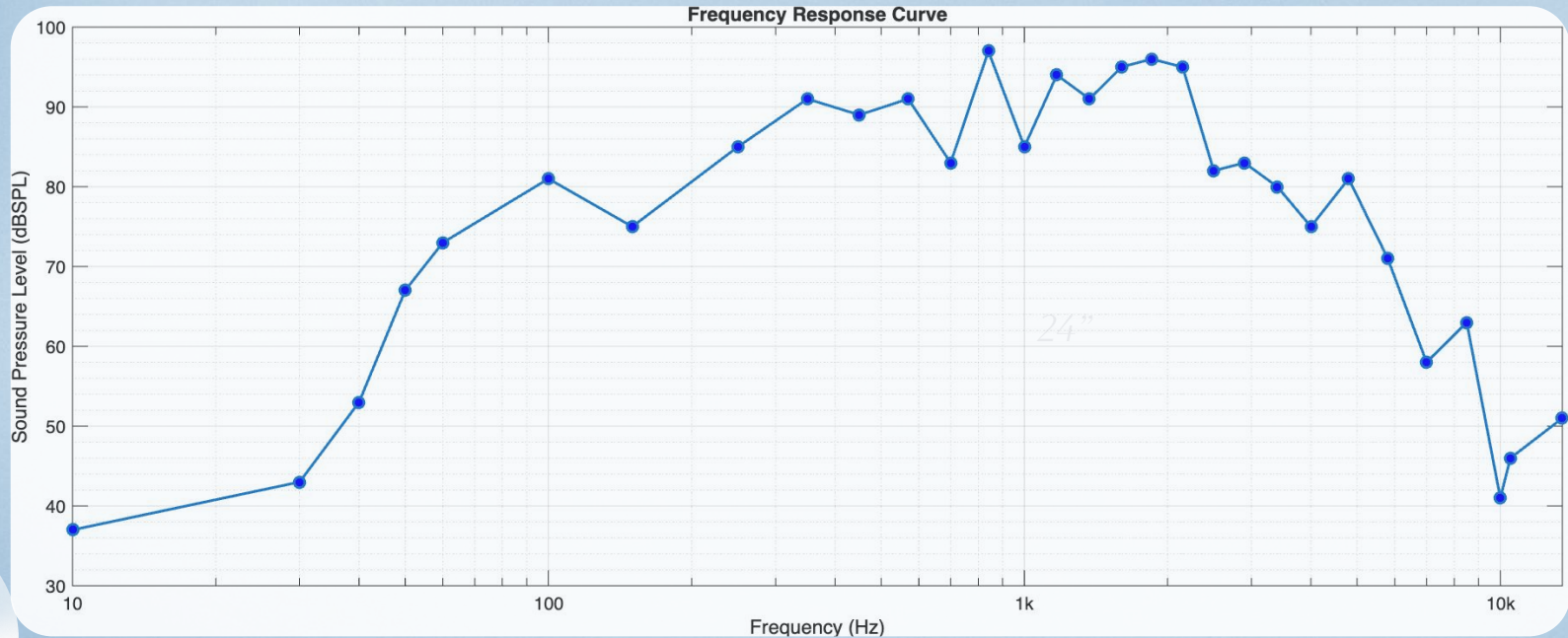
ENCLOSURE BUILD



Rhino model and build by Lorenzo Lanaro

MEASUREMENTS

Below is the measurements of SPL at: 10, 100, 1000, and 10,000 Hz, and bark bands @ 1m

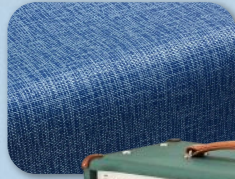


FUTURE WORK

TOLEX & HANDLE

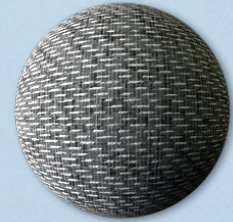


Covering the amp in tolex would add a nice texture & protection. I wasn't a fan of the handles I bought.



PROTECTION

Add a frame with protective cloth on the inner border of the amp face.



REINFORCEMENT

The corners can be reinforced with leather or metal amp angles if necessary.



TWEETER?

The crossover I have has an extra panel for a bass amp horn, but this would have to be gain staged much lower than the bass driver, but this is not required, it sounds good now.



FINAL BUILD



Made with assistance from Sam Courtney

THANKS!

Let's listen to the amp!

Any questions?
Ask away!

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